Program Committee
CSCWD 2027

Re: *Evidence Routing and Supported Residuals for Cooperative LiDAR–Radar Perception*

Dear Program Committee,

Please consider the manuscript “Evidence Routing and Supported Residuals for Cooperative LiDAR–Radar Perception” for CSCWD 2027. The paper introduces evidence routing and supported residual correction for local LiDAR–radar features before cooperative aggregation, where missing returns and spurious echoes require different responses.

DEER uses independently supervised missing/spurious scores to guide feature routing while retaining a positive gain for each available modality. SBR selects nearby measured radar donors, allows donor rejection, and constrains the added residual by measured support and current feature amplitude. Both modules precede the shared cooperative backbone; detection supervision learns the correction direction without a feature teacher.

Seven configurations use one seed, 20 training epochs, and model selection on Normal/Fog validation data in the simulated V2X-R dataset. The predetermined joint model AB achieves 49.19% Snow Vehicle AP3D at IoU 0.7, a gain of 3.52 percentage points over the internal baseline. The Fog/Snow mean increases by 1.67 points, while Normal/Fog decrease by 0.26/0.19 points. The multiplicative type-prior control N2 achieves the higher weather mean, 50.62% versus AB’s 50.13%. These findings identify strict-overlap Snow detection as the strongest benefit of the joint design.

The study fits the conference’s interests in sensor networks and collaboration technology applications by examining the local feature processing that supplies a multi-agent perception system. The study separates degradation evidence from the support and magnitude of local corrections.

Funding: To be confirmed with the authors.

Thank you for considering this manuscript.

Sincerely,

Anonymous Authors